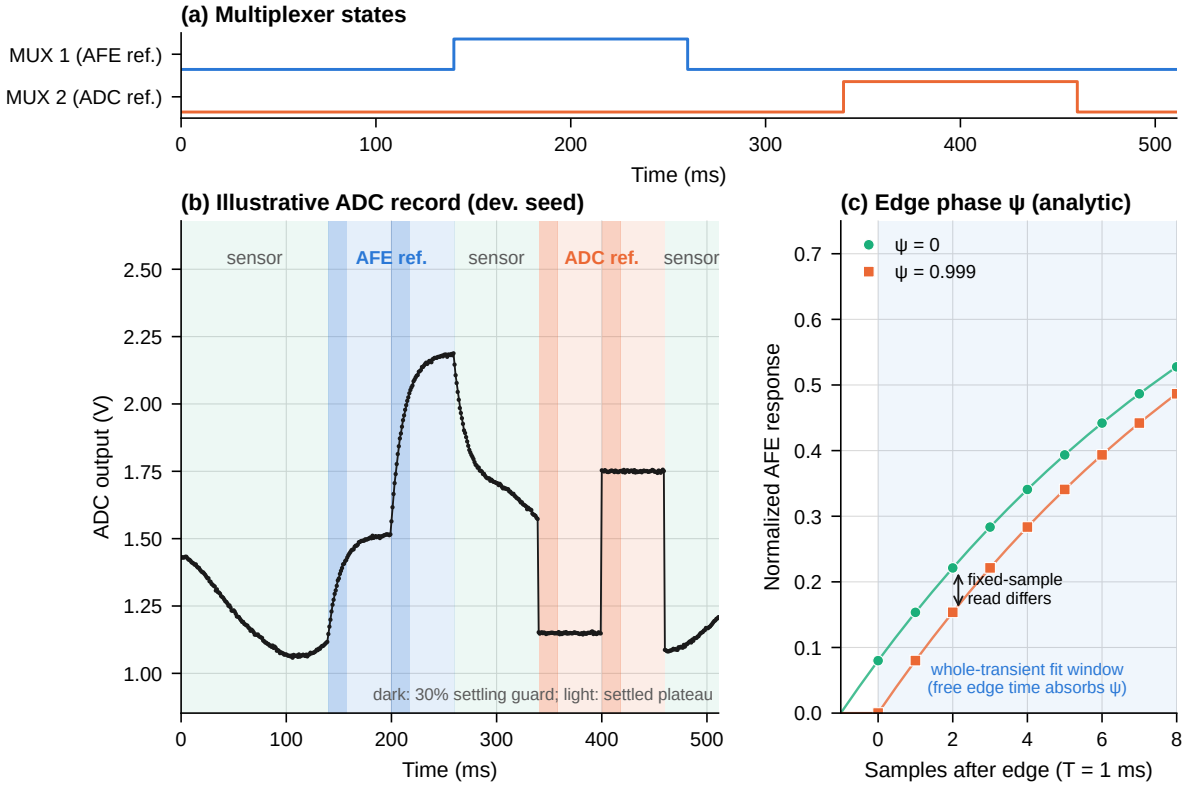
(a) Multiplexer states
MUX 1 (AFE ref.)
MUX 2 (ADC ref.)
0
100
200
300
400
500
Time (ms)
(b) Illustrative ADC record (dev. seed)
sensor
AFE ref.
sensor
ADC ref.
sensor
ADC output (V)
2.50
2.25
2.00
1.75
1.50
1.25
1.00
dark: 30% settling guard; light: settled plateau
Time (ms)
(c) Edge phase ψ (analytic)
ψ = 0
ψ = 0.999
Normalized AFE response
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
fixed-sample
read differs
whole-transient fit window
(free edge time absorbs ψ)
0
2
4
6
8
Samples after edge (T = 1 ms)